

Cloud Computing CET322

Introduction to Cloud Computing

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Class Warm Up and Expected Learning Outcomes

- Are you familiar with Linux?
- Are you familiar with scripting using Python or Bash?
- Are you familiar with Databases and have you written SQL statements before?
- Do you have a background in computer architecture, OS, and Networking?
- Have you written a web service before?
- Are you comfortable developing web services in Python, Java, or JavaScript?

Quick Response!

CET322: Class Warm up!



CET322: Course Syllabus & Office Hours

CET322: Cloud Computing

Contact Time				CH	SWL	Exams Time		Marks		
LCT	TUT	LAB	Total			MT	FE	CW	ME	FE
120	0	120	240	3	120	90	120	60	15	25

- Requisites
 - Prerequisites: CET212 (Operating Systems), AND CET161 (Network Basics)
- Course Description:
 - This course introduces the principles and practices of cloud computing as the on-demand delivery of IT resources over the Internet.
 - Students will study cloud service models (SaaS, PaaS, IaaS, IDaaS), virtualization, cloud storage, collaboration, security, disaster recovery, business continuity, and service-oriented architecture.
 - The course also covers managing, governing, and migrating to the cloud, along with designing cloud-based solutions and case studies. Laboratory sessions provide hands-on experience with cloud concepts and tools.



CET322: Cloud Computing



Credit Hours

3



Contact Time

Lecture: 120 min • Lab: 120 min
Total: 240 min



Student Workload

120 min (self-study
& assignments)



Exams

Midterm: 90 min • Final: 120 min



Assessment

Coursework: 60% • Midterm Exam: 15%
Final Exam: 25%

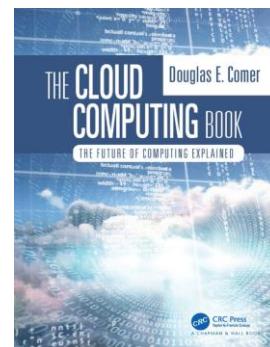
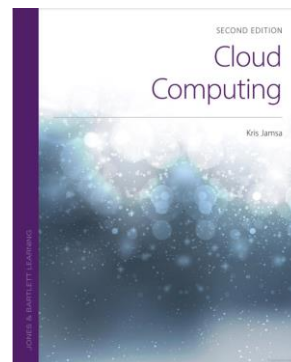
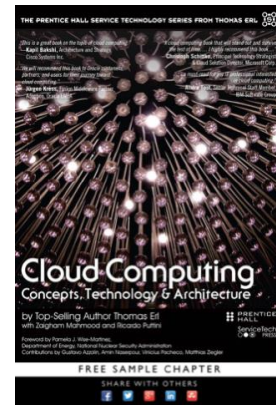
Learning Resources and References

❏ Books:

- ***Cloud Computing: Concepts, Technology & Architecture*** by Thomas Erl
- ***The Cloud Computing Book, The Future Of Computing Explained*** by DOUGLAS E. COMER
- ***Cloud computing, 2nd Edition***, by Jamsa, K., O'Reilly, Jones & Bartlett Publishers

❏ Cloud Platforms Offering Learning Resources:

- Amazon Web Services (AWS) – AWS Academy / AWS Educate
 - URL: <https://aws.amazon.com/training/>
- Microsoft Azure – Azure for Students / Microsoft Learn
 - URL: <https://learn.microsoft.com/en-us/training/>
- Google Cloud Platform (GCP) – Google Cloud Skills Boost
 - URL: <https://cloudskillsboost.google/>
- IBM Cloud – IBM SkillsBuild / IBM Cloud Learning Hub
 - URL: <https://www.ibm.com/training/cloud>
- Oracle Cloud – Oracle Academy & Learning
 - URL: <https://learn.oracle.com/ols/home>



CET322: Cloud Computing Course Goals

Description: Students gain hands-on experience solving real world problems by completing projects in the areas of **cloud computing**, which utilize existing public cloud tools and services. Students are exposed to real-world data, scenarios, infrastructure and budgets in order to learn how to:

1. **Design**, architect, implement, test, deploy, monitor and maintain cloud-based applications;
2. **Identify** the appropriate tools and architectures to implement a cloud-based design;
3. **Analyze** the tradeoffs between different tools and cloud offerings to meet real-world constraints;
4. **Evaluate** performance characteristics of cloud-based services to implement optimizations;
5. **Collaborate** with a team on an open-ended project to incrementally realize an optimized end-to-end cloud-based solution.

CET322: Cloud Computing Course Learning Outcomes

- **CLO 1:** Analyze and implement the cloud deployment and service models to solve the computing problems per the given situation.
- **CLO 2:** Evaluate and derive various cloud delivery models and security levels for cloud-based infrastructure to accomplish user requirements.
- **CLO 3:** Demonstrate the ability to accomplish group tasks related to designing the cloud-based solution for the required infrastructure of the organization.
- **CLO 4:** Formulate system goals and requirements, data privacy requirements, capacity planning and scaling capabilities for the cloud infrastructure to communicate effectively to all stakeholders.
- **CLO 5:** Apply the security techniques, perform experiments, interpret data and draw results and conclusions.
- **CLO 6:** Write lab reports and present results.

CET322: Course Tutorials

Tutorials:

- WEEK 1: Lab **Introduction**, devices, simulator, Course Policies, reports
- WEEK 2: Install Oracle Virtual box and create two **virtual machines** on the computer.
- WEEK 3: Install a **C/C++ compiler** in the virtual machine created using virtual box and execute Simple Programs
- WEEK 4: Install **Google App Engine**. Create hello world app and other simple web applications using python/java
- WEEK 5: Test ping command to test the communication between the **guest OS and Host OS**
- WEEK 6: Simulate a cloud scenario using **CloudSim** and run a scheduling algorithm that is not present in CloudSim
- WEEK 7: Find a procedure to **transfer** the **files** from one virtual machine to another virtual machine.
- WEEK 8: Install **Hadoop single node cluster** and run simple applications.
- WEEK 9: Develop **hadoop application** to count no of characters, no of words and each character frequency.
- WEEK 10: Develop **hadoop application** to process given data and produce results such as finding the year of maximum usage, year of minimum usage.
- WEEK 11: Establish an **AWS account**. Use the AWS Management Console to launch an EC2 instance and connect to it.
- WEEK 12: Design a protocol and **use Simple Queue Service(SQS)** to implement the barrier synchronization after the first phase
- WEEK 13: Develop a Hello World application using **Google App Engine**. Write a Google app engine program to generate n even numbers and deploy it to Google cloud
- WEEK 14: Develop a Guestbook Application using **Google App Engine**, Google app engine program multiply two matrices
- WEEK 15: A) Develop a **Windows Azure** Hello World application using. B) Google app engine program to validate the user use **mysql** to store user info and deploy on to cloud , also implement it with Azure.

CET322: mini Projects

1. Installing Linux
2. Installing Virtual Machine
3. Setting Cloud Storage
4. Deploying a Web application



Group Projects on AWS/Azure/GCP Clouds – at Least One Project per Group

o. AWS/Azure/GCP Account Setup & Data Analytics

- Benchmarking VMs, SSH, Authentication, Billing, Security Groups, Vertical Scaling

1. Scaling, Elasticity and Failure with VMs

- Auto Scaling, Load Balancing, Monitoring

2. Cloud Elasticity with Containers

- Docker Containers, Kubernetes

3. Cloud Storage

- Standalone MySQL, Azure Managed SQL DB

4. Analytics Engines for the Cloud

- Spark, Cloud ML Frameworks

Team Project: A Cloud Native Web Service

- No restrictions on tools or AWS services
- Evaluated based on cost and performance

Labs and Cloud Subscriptions

❖ Microsoft Azure



Choose Azure if you're **already invested in Microsoft** (AD/Entra, Windows Server, SQL Server) or you need robust **hybrid** and compliance governance out of the box.

❖ Google Cloud Platform



Choose GCP if your priority is **analytics/AI and Kubernetes**, with straightforward billing that rewards steady usage automatically.

Google Cloud Platform

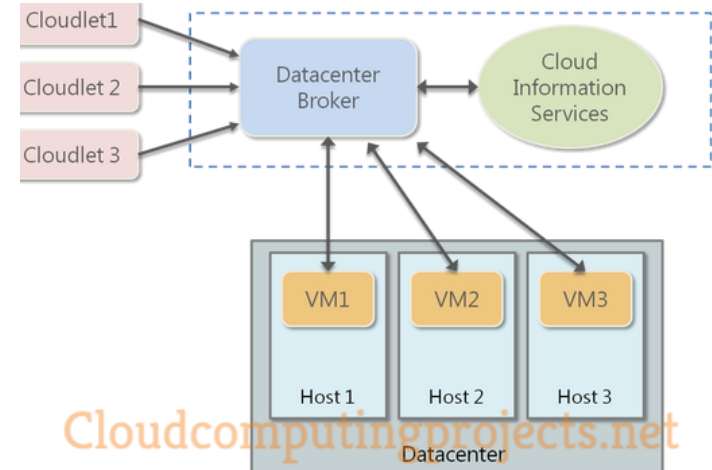
❖ Amazon Web Services



Choose AWS if you need the **widest set of building blocks**, global scale, and lots of cost levers to optimize heterogeneous workloads.

CloudSim

- CloudSim is an open-source simulation toolkit written in Java for modeling and evaluating cloud computing infrastructures and services before real-world deployment.
- It provides classes to describe data centers, virtual machines, and applications, enabling researchers to test new algorithms and policies for resource provisioning, scheduling, and management.
- CloudSim supports both single and federated cloud environments, allowing for the simulation of network configurations, workloads, and resource performance.



A cloudlet in cloud computing is a small-scale, trusted data center located at the edge of the internet, designed to provide cloud services to nearby mobile devices and applications.

To summarize



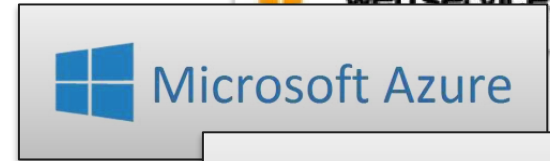
CET322: Expected Learning Outcomes!

**So What is Cloud
Computing?**

What is Cloud Computing?

The US National Institute of Standards and Technology (**NIST**) **Definition**:

“A model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.”

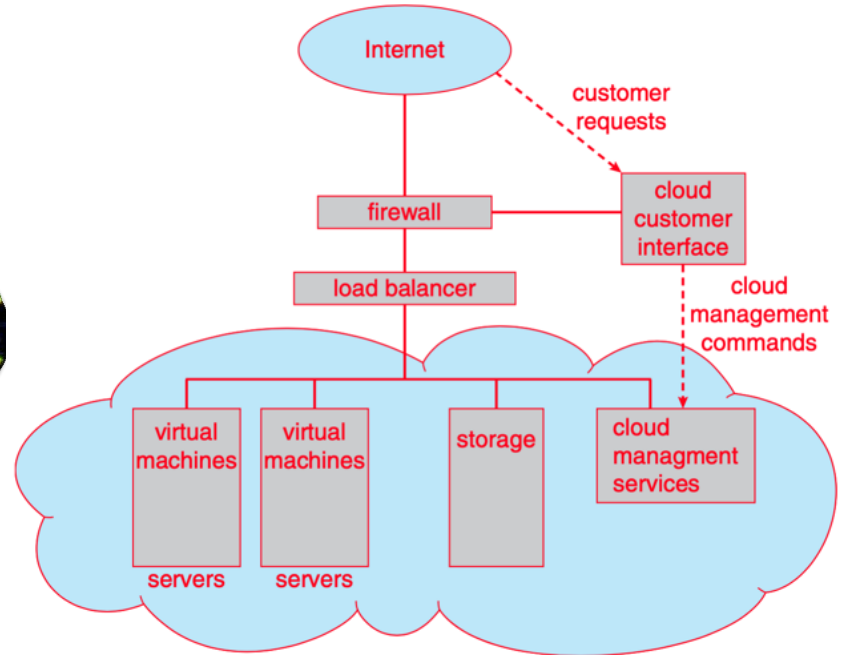


Cloud Computing Concept

- **“Cloud computing** is an [information technology](#) (IT) paradigm that enables ubiquitous access to shared pools of configurable [system resources](#) and higher-level services that can be rapidly [provisioned](#) with minimal management effort, often over the [Internet](#).
- Cloud computing relies on sharing of resources to achieve coherence and [economies of scale](#), similar to a [public utility](#).” https://en.wikipedia.org/wiki/Cloud_computing
- “Simply put, cloud computing is the delivery of computing services – servers, storage, databases, networking, software, analytics and more – over the Internet (“the cloud”).
- Companies offering these computing services are called cloud providers and typically charge for cloud computing services based on usage, similar to how you're billed for gas or electricity at home.” <https://azure.microsoft.com/en-gb/overview/what-is-cloud-computing/>

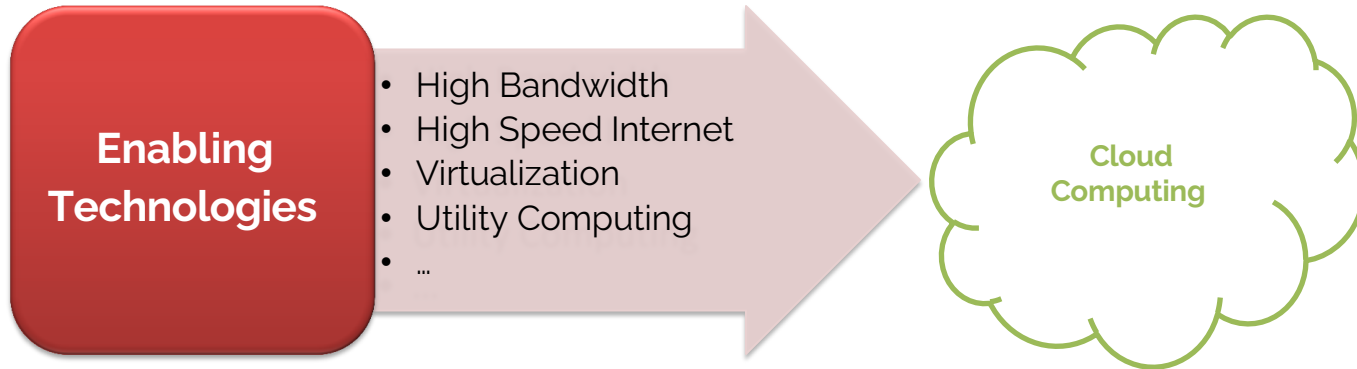
A Cloud is ...

- ❑ Datacenter hardware and software that the vendors use to offer the computing resources and services



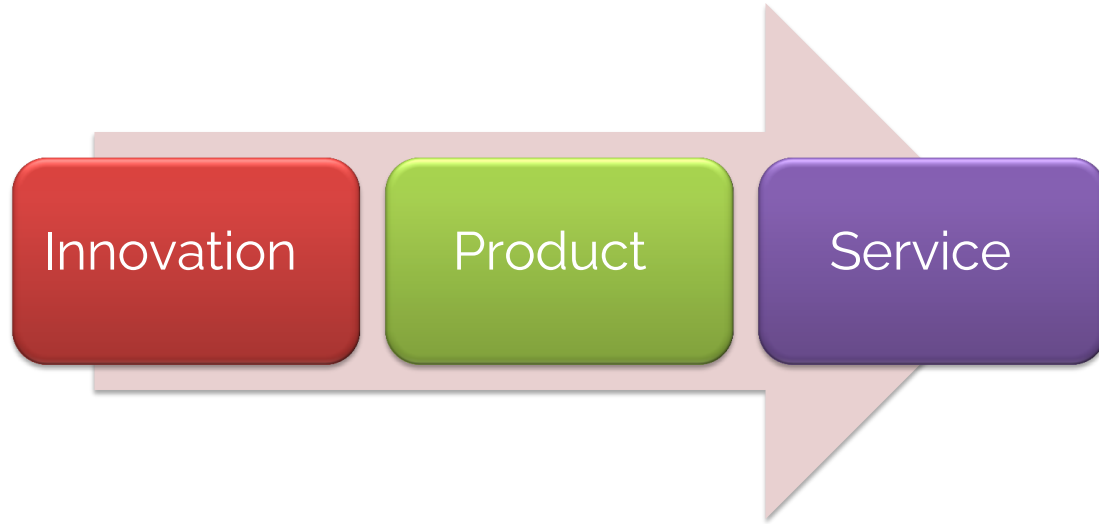
Cloud Computing and the Internet

- + Think of it as Internet Computing
 - + Computation done over the Internet (Graphics, Gaming, etc.)

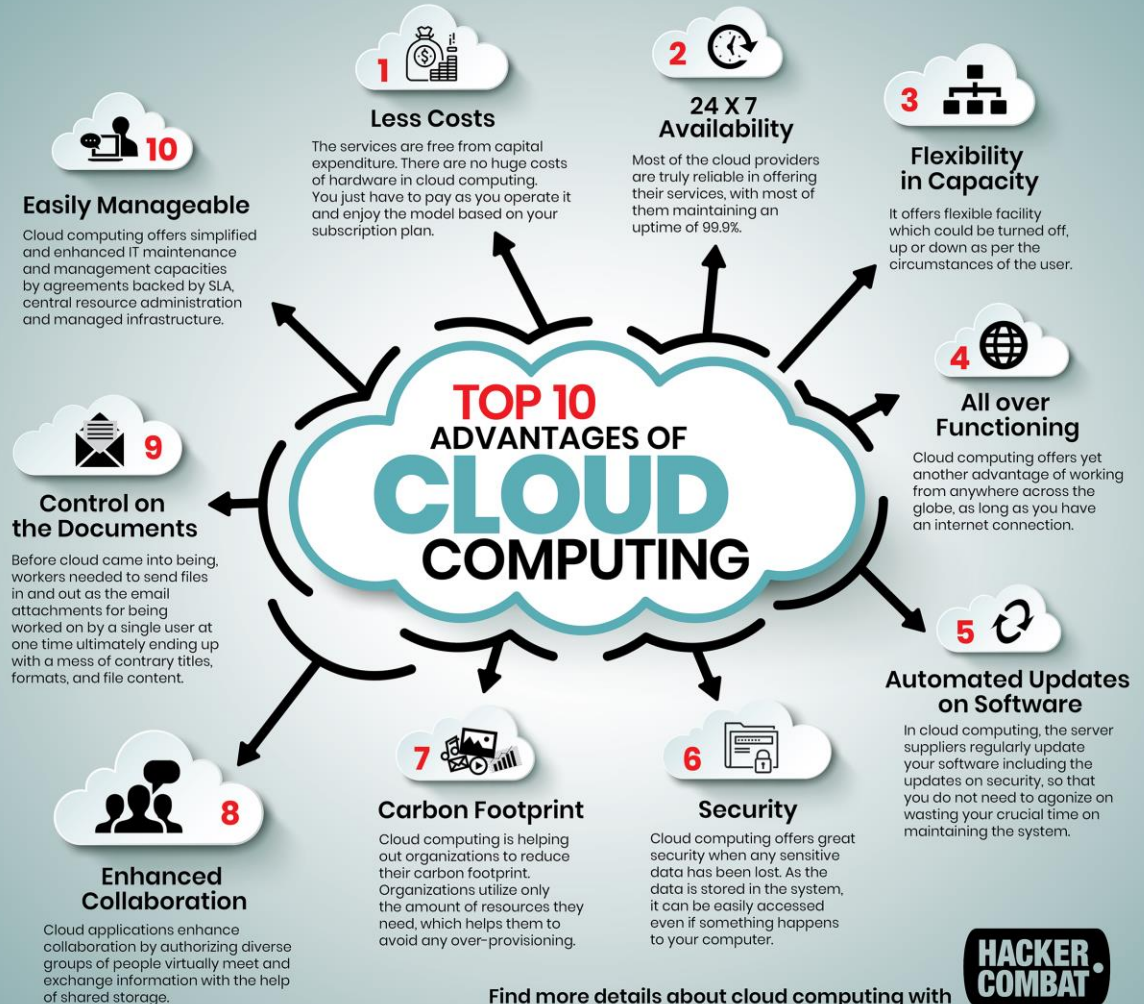


Can We Define Cloud Computing?

*"Cloud Computing is the transformation of IT
from a product to a service"*



Advantages of Cloud Computing



Find more details about cloud computing with

**HACKER
COMBAT**

Disadvantages of Cloud Computing

Disadvantages of Cloud Computing

Data Security and
Privacy Concerns



Hidden Costs and
Budget Overruns



Vendor
Lock-In Risks



Downtime and
Reliability Issues



Limited Customization
and Control

Components of the Cloud

- Front-end (regular desktop, thin client, mobile device)
- Back-end (servers)
- Storage / Datacenters
- Delivery Service
(SaaS, PaaS, IaaS)



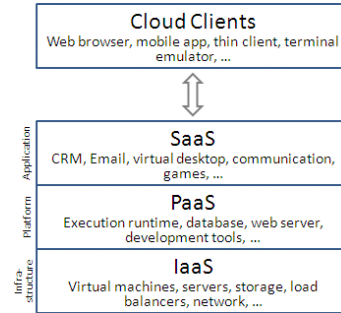
Cloud Computing Stack

- Applications
- Development Platforms
- Elasticity
 - APIs to enable automation, Alarms, protocols, triggers, etc...
- Sharing mechanisms
 - Virtualization, Containers, ...
- Distributed systems
 - Programming models
 - Storage
- Data centers

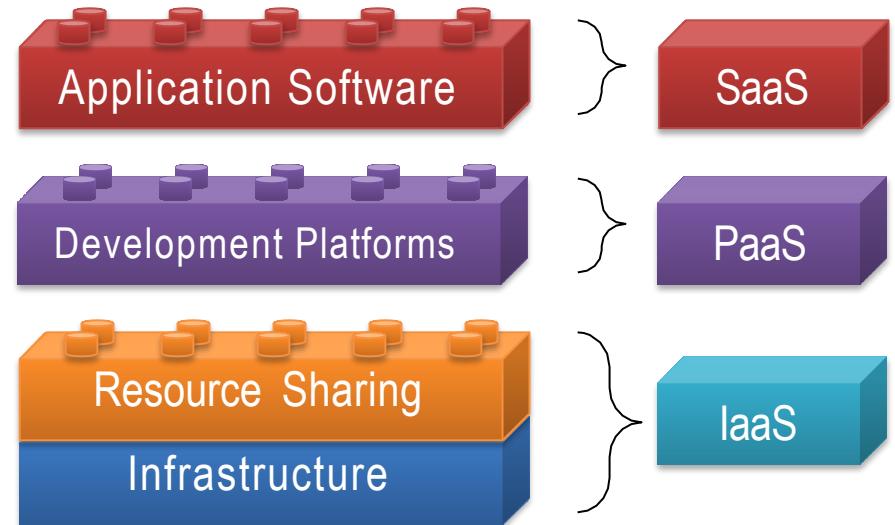


Cloud Building Blocks and Delivery Models

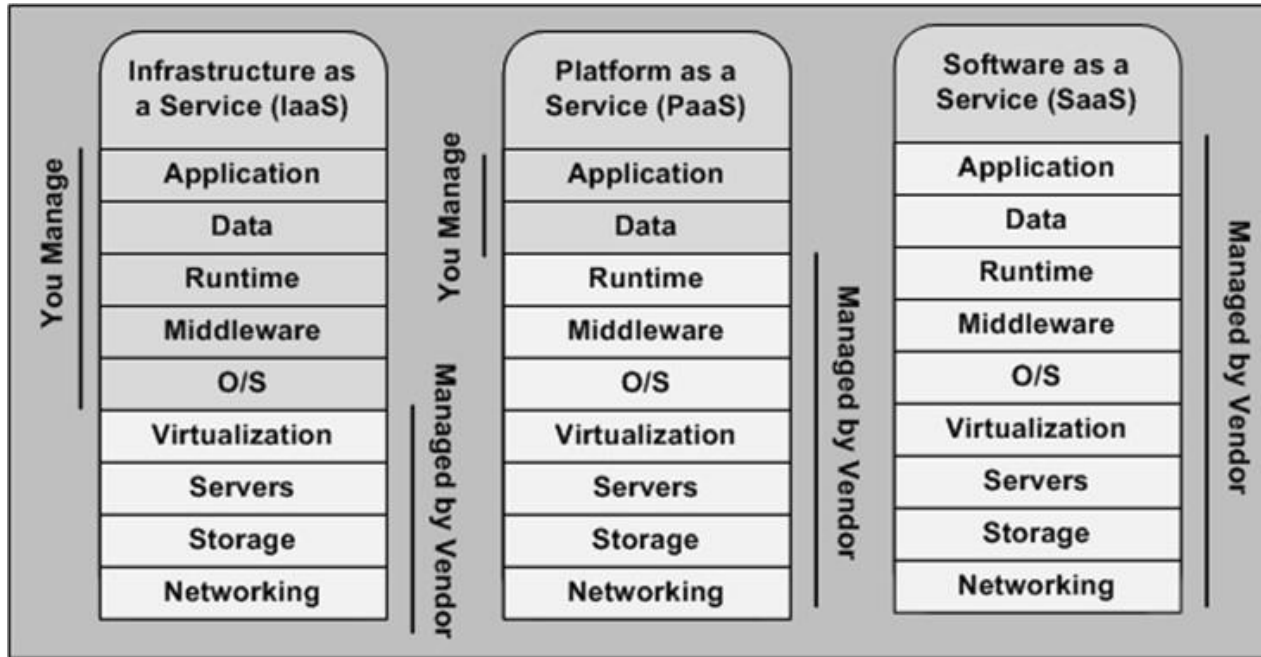
Cloud services are available in various forms, corresponding to the layer of abstraction desired by the user



- Software as a Service (**SaaS**)
(High Level)
- Platform as a Service (**PaaS**)
- Infrastructure as a Service (**IaaS**)
(Low Level)



The Three delivery models of Cloud Computing: IaaS, PaaS, SaaS



Software as a Service (SaaS)

- Software is delivered through the internet over a browser or mobile application
- Replace desktop software with cloud-based versions
- Webmail, Productivity Software, ERP, CRM etc.
- Centrally managed, globally available, automatically updated



Adobe® Creative Cloud



Platform as a Service (PaaS)

- Tools and APIs to develop and deploy cloud-based applications
- Create customized SaaS in the form of Web or mobile applications



Googles	AppEngine is a development platform based upon Python and Java
force.com	Apex a development platform based upon a proprietary programming language
Microsoft	Azure provides a development platform based upon .Net.



Infrastructure as a Service (IaaS)

- Compute, storage and network resources bundled in the form of virtual machines
- The user is able to deploy and run arbitrary software, which can include operating systems and applications.
- The user does not manage or control the underlying Cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of some networking components, e.g., host firewalls.
- Fully flexible in terms of software and environment

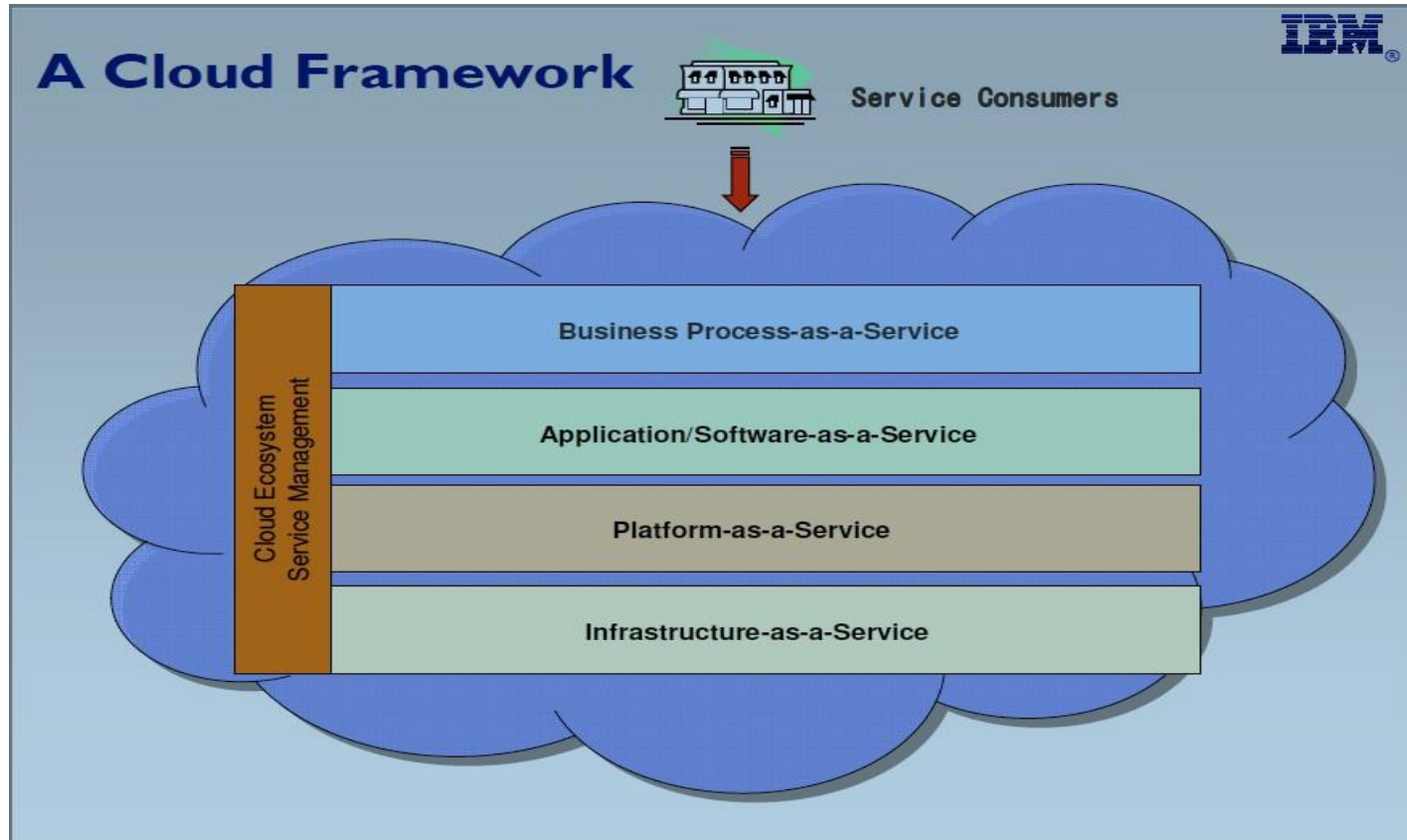


More Service Models

- DaaS – Data as a Service
Consumer queries against provider's database
- NaaS – Network as a Service
Provider offers virtualized networks (e.g. VPNs)



Everything-as-a-Service



Cloud Marketplace



Cloud Broker Platform



Cloud Management



SaaS, PaaS,
and IaaS



Cloud Platform



Virtualization Software/Mgmt



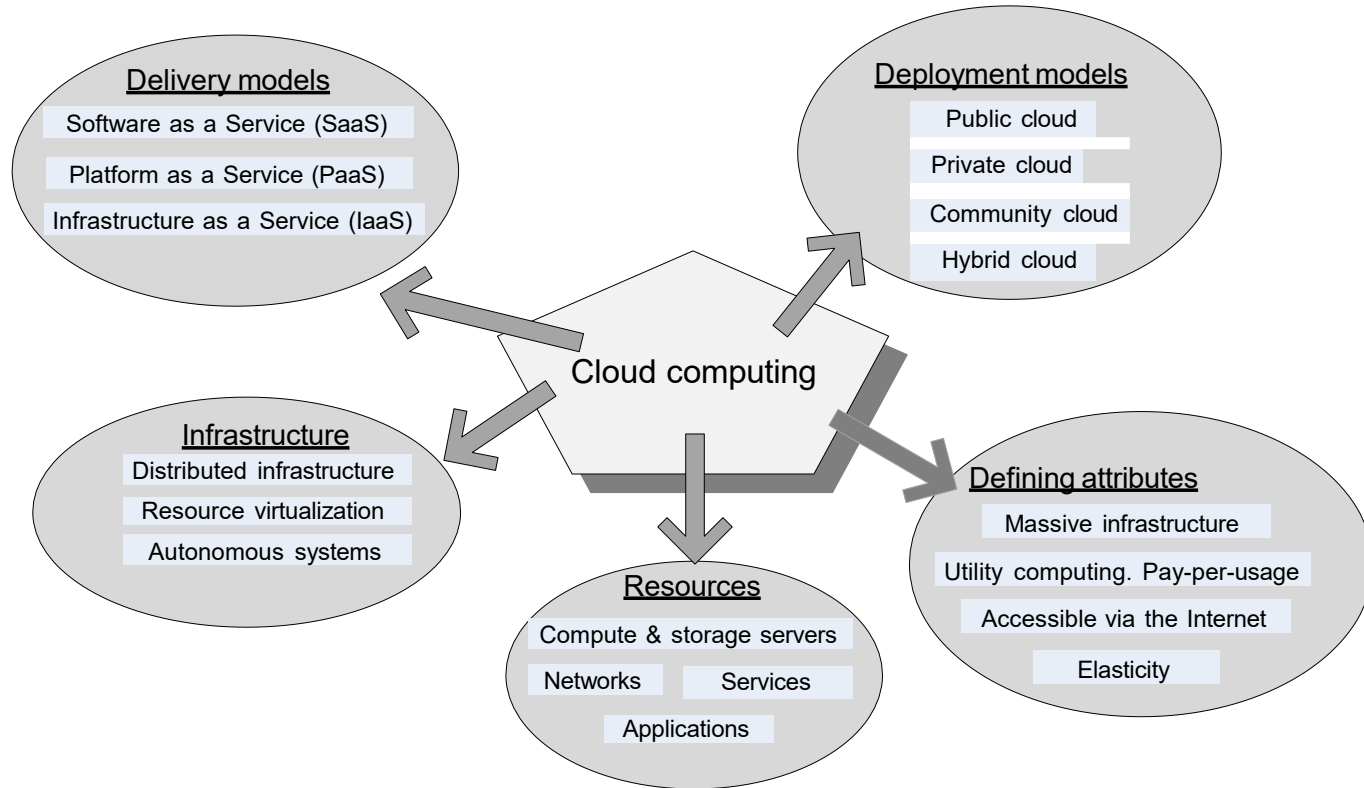
Hardware



Types of clouds

1. **Public Cloud** - the infrastructure is made available to the general public or a large industry group and is owned by the organization selling cloud services.
2. **Private Cloud** - the infrastructure is operated solely for an organization.
3. **Hybrid Cloud** - composition of two or more Clouds (public, private, or community) as unique entities but bound by a standardised technology that enables data and application portability.
4. **Other types: e.g., Community/Federated Cloud** - the infrastructure is shared by several organizations and supports a community that has shared concerns.

Cloud Computing Models, Resources, Attributes



Privacy issues

- Cloud service providers have already collected petabytes of sensitive personal information stored in data centers around the world. The acceptance of Cloud Computing therefore will be determined by privacy issues addressed by these companies and the countries where the data centers are located.
- Privacy is affected by cultural differences; some cultures favour privacy, others emphasise community. This leads to an ambivalent attitude towards privacy in the Internet which is a global system.

Cloud Vulnerabilities

- Clouds are affected by malicious attacks and failures of the infrastructure, e.g., power failures.
- Such events can affect the Internet domain name servers and prevent access to a Cloud or can directly affect the Clouds:
 - in 2004 an attack at Akamai caused a domain name outage and a major blackout that affected Google, Yahoo, and other sites.
 - in 2009, Google was the target of a denial of service attack which took down Google News and Gmail for several days;
 - in 2012 lightning caused a prolonged down time at Amazon.

Early Models of Cloud Computing

- Basic reasoning: information and data processing can be done more efficiently on large farms of computing and storage systems accessible via the Internet.
- Two early models:
 1. **Grid computing** – initiated by the National Labs in the early 1990s; targeted primarily at scientific computing.
 - “Grid computing is the collection of computer resources from multiple locations to reach a common goal. The grid can be thought of as a distributed system with non-interactive workloads that involve a large number of files.” from Wikipedia
 2. **Utility computing** – initiated in 2005-2006 by IT companies and targeted at enterprise computing.
 - “Utility computing is a service provisioning model in which a service provider makes computing resources and infrastructure management available to the customer as needed, and charges them for specific usage rather than a flat rate.” from Wikipedia

Back to Basics -- Parallel Computing

- **“Parallel computing** is a form of computation in which many calculations are carried out simultaneously, operating on the principles that large problems can often be divided into smaller ones, which are then solved concurrently (in parallel).” Wikipedia
- Hardware and software systems allow us to:
 - Solve problems demanding resources not available on a single system.
 - **Reduce the time required to obtain a solution.**

Back to Basics -- Distributed systems

- Collection of autonomous computers, connected through a network and distribution software (often) called middleware which enables computers to coordinate their activities and to share system resources for a common goal.
- Characteristics:
 1. The users perceive the system as a single, integrated computing facility.
 2. The components are autonomous.
 3. Scheduling and other resource management and security policies are implemented by each system.
 4. There are multiple points of control and multiple points of failure.
 5. The resources may not be accessible at all times.
 6. Can be scaled by adding additional resources.
 7. Can be designed to maintain availability even at low levels of hardware/software/network reliability.

Parallel Computing – Amdahl's Law

- The speedup S measures the effectiveness of parallelisation:

$$S(N) = T(1) / T(N)$$

- $T(1)$ □ the execution time of the sequential computation.
 - $T(N)$ □ the execution time when N parallel computations are executed
-
- **Amdahl's Law:** if α is the fraction of running time a sequential program spends on non-parallelisable segments of the computation then:

$$S \Rightarrow 1 / \alpha$$

- This is a theoretical upper bound on the best speedup we can get from parallelising a certain program.

Example on Amdahl's Law

- Amdahl's Law is a principle in computer architecture and parallel computing that explains the theoretical maximum speedup of a program when part of it is parallelized.
- Key Idea
 - If a large portion of the program is serial, adding more processors gives little benefit.
 - If most of the program is parallelizable, speedup increases significantly — but only up to a limit.

Formula

$$S(N) = \frac{1}{(1 - P) + \frac{P}{N}}$$

Where:

- $S(N)$ = speedup with N processors
- P = fraction of the program that can be parallelized
- $(1 - P)$ = fraction that must remain serial (non-parallel)
- N = number of processors

Example

Suppose:

- $P = 0.9$ (90% parallelizable)
- $N = 10$ processors

$$S(10) = \frac{1}{(1 - 0.9) + \frac{0.9}{10}} = \frac{1}{0.1 + 0.09} = \frac{1}{0.19} \approx 5.26$$

Even with **10 processors**, the maximum speedup is only ~5.26x, not 10x, because of the **serial bottleneck**.

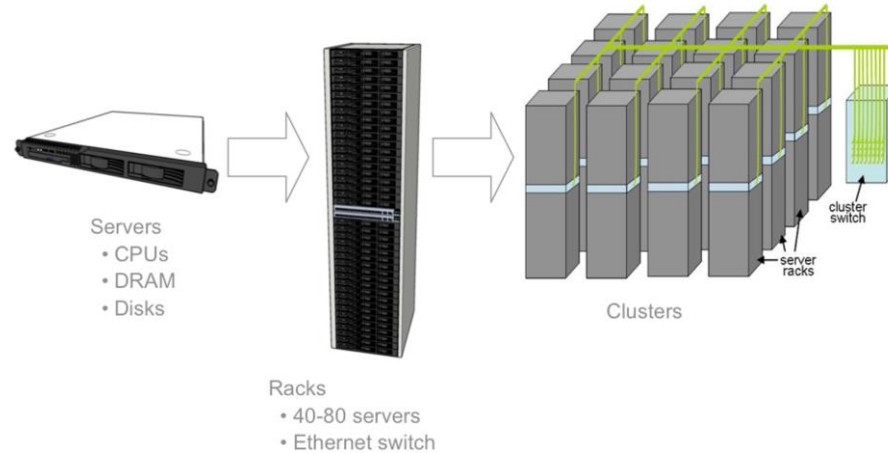
Performance metrics

- Metrics (database server as an example)
 - Throughput: I/O operations/second, data transfer rate/second
 - This is system view
 - If throughput is low, more servers used than really needed
 - Latency: seek time (seconds), response time (seconds)
 - This is user view
 - If latency is high, users have slow response
 - Utilization: percentage of data transfer rate, disk capacity used
 - This is future view
 - If avg utilization > 60%, your systems will crawl on busy days
 - Average server utilization ~15-20% in traditional data centers
- Units
 - **Ethernet data transfer:** MB/sec
 - **Disk data transfer:** MB/sec
- Use comparable units when assessing systems
 - 8 bits (b)= 1 byte (B)
 - MB, GB, TB

Cloud Computing Infrastructure and VMWARE



Datacenter Hardware



- ❑ Rows of rack-mounted servers
- ❑ Datacenter: 50 – 200K of servers, 10 – 100MW
Often organized as few and mostly independent clusters

Datacenter Example



Datacenter HW: Compute

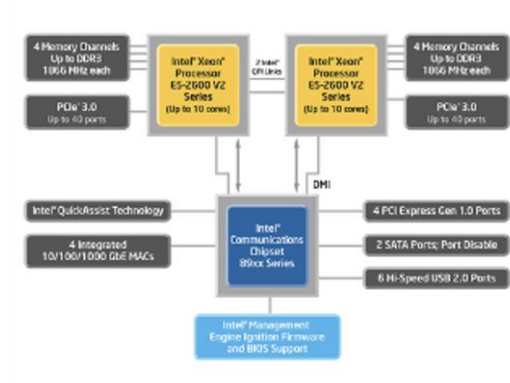
The basics

Multi-core CPU servers 1 & 2 sockets

What's new

- GPUs
- Custom accelerators (AI) FPGAs

2-socket server



Nvidia GPU



Google TPU²



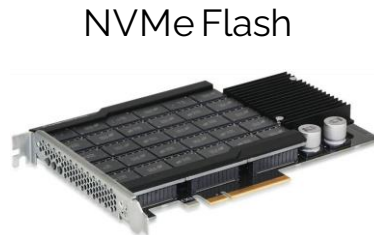
Microsoft Catapult



Datacenter HW: Storage

The basics

- Disk trays
- SSD & NVM Flash



NVMe Flash

JBOD disk array

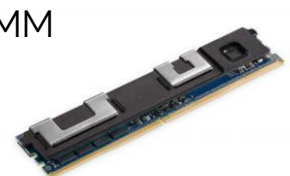


JBOD
Just a Banch
of Disks

What's new

- Non-volatile memories
- New archival storage (e.g., glass)

NVM DIMM



Distributed with compute or NAS systems

- Remote storage access for many use cases (why?)

Datacenter HW: Networking

- The basics

- 40, 100, 200 GbE NICs
- 100GbE to 200 GbE switches
- Clos topologies

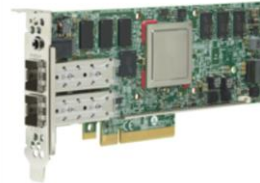
100GbE Switch



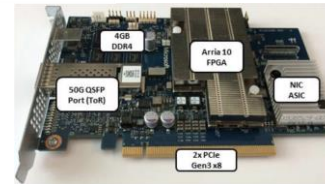
- What's new

- Software defined networking
- In network computation
- Smart NICs
- FPGAs

Smart NIC



Microsoft Catapult



Performance Metrics

Throughput

- Requests per second
- Concurrent users
- Gbytes/sec processed

...

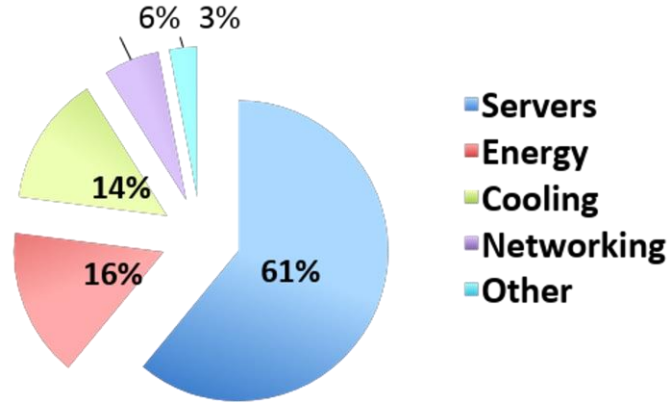
Latency

- Execution time
- Per request latency

Total Cost of Ownership (TCO)

- **TCO = capital (CapEx) + operational (OpEx) expenses**
- Operators perspective
 - CapEx: building, generators, A/C, compute/storage/net HW Including spares, amortized over 3 – 15 years
 - OpEx: electricity (5-7c/KWh), repairs, people, WAN, insurance, ...
- Users perspective
 - CapEx: cost of long term leases on HW and services OpEx: pay per use cost on HW and services, people

Operator's TCO Example



[Source: James Hamilton]

Hardware dominates TCO, make it cheap Must utilize it as well as possible

Example of Cloud-enabling Data Centers

- Large warehouse scale data centers
- Growing at a rapid rate
- Next is an example from Microsoft Azure
 - Azure US-East2 (Boydton, VA)
 - Azure Expansion 1
 - Azure Expansion 2



US-East2
(⁵³Boydton, VA)

Reliability

- Failure in time (FIT)

Failures per billion hours of operation = $10^9/\text{MTTF}$

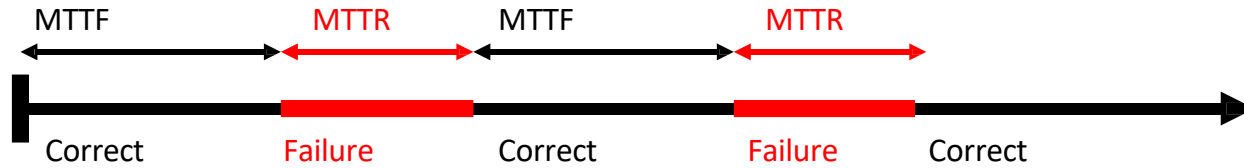
- Mean time to failure (MTTF)

Time to produce first incorrect output

- Mean time to repair (MTTR)

Time to detect and repair a failure

Availability



$$\text{Steady state availability} = \text{MTTF} / (\text{MTTF} + \text{MTTR})$$

Key Availability Techniques

Technique	Performance	Availability
Replication	✓	✓
Partitioning (sharding)	✓	✓
Load-balancing	✓	
Watchdog timers		✓
Integrity checks		✓
Canaries		✓
Eventual consistency	✓	✓

Make apps do something reasonable when not all is right

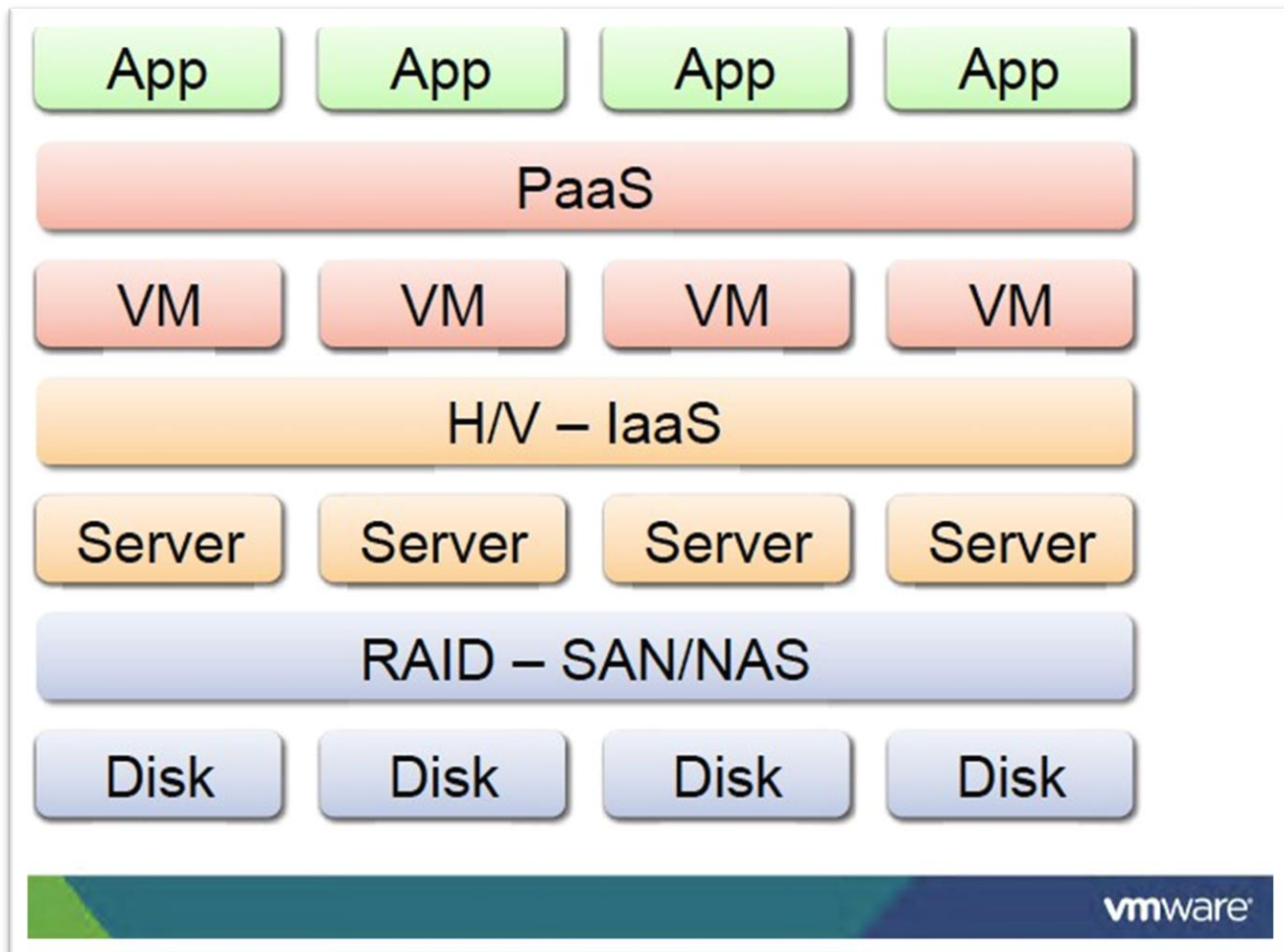
Better to give users limited functionality than an error page

Aggressive load balancing or request dropping

Better to satisfy 80% of the users rather than none

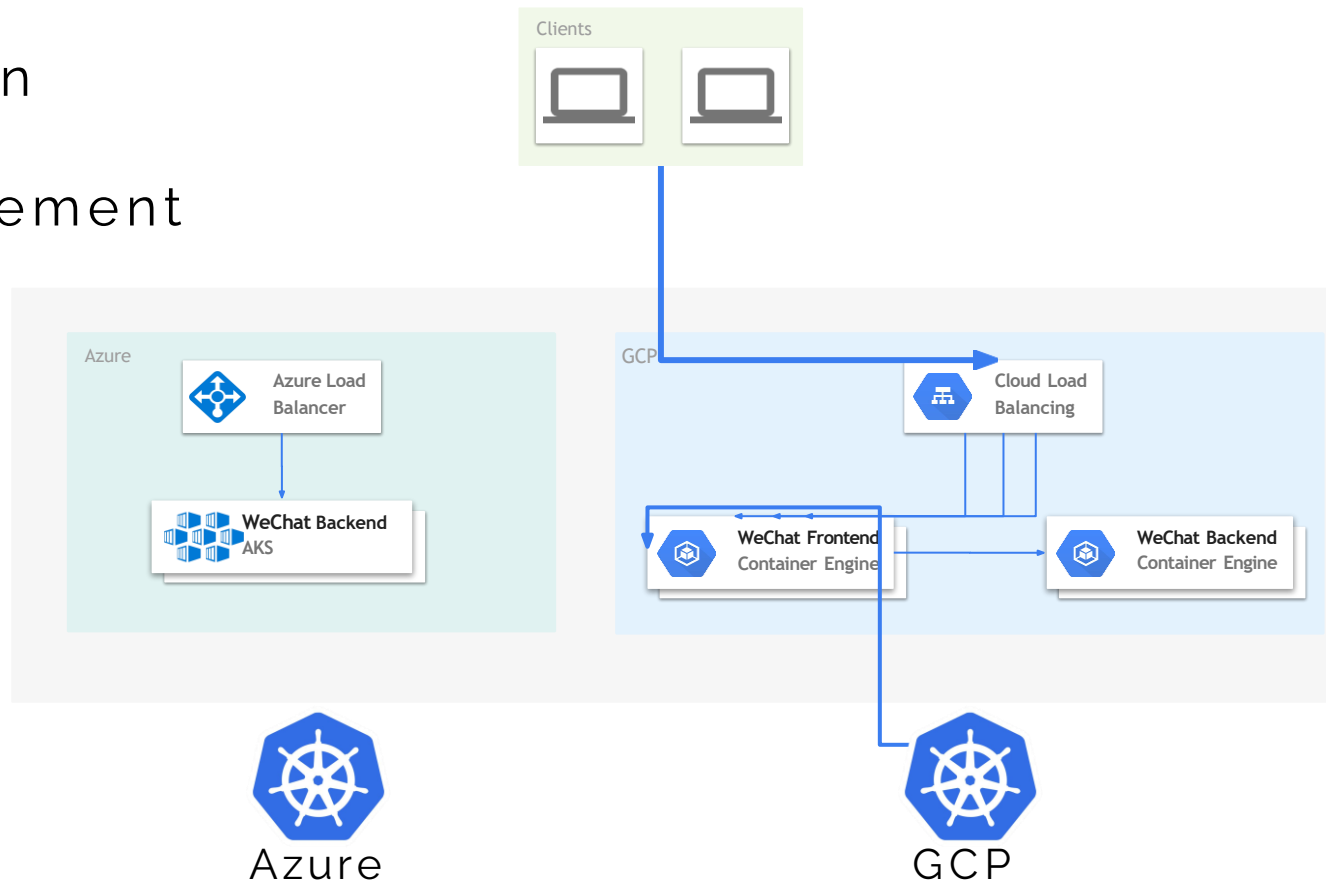
VMWARE STACK

- **RAID**
(Redundant Array of Independent Disks).
- **SAN**
(Storage Area Network)
- **NAS**
(Network Attached Storage)



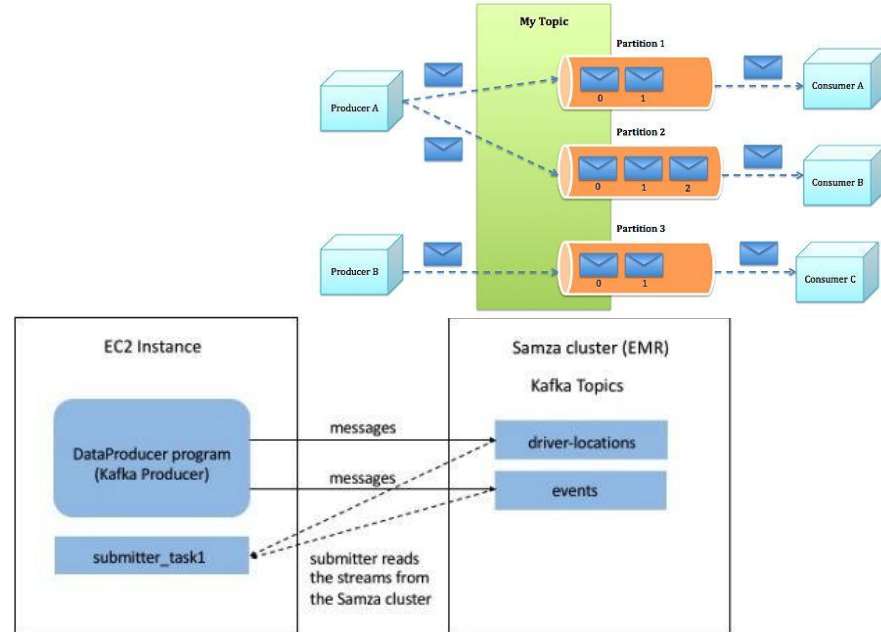
Containers and Kubernetes

- Containerization
- Cluster Management
- Multi-Cloud Deployment



Uber-like Application for Cairo

- Stream Processing with Kafka/Samza
 - Stream 1: Car GPS coordinates
 - Stream 2: Customers
- Task:
 - Match customers with cars to minimize travel time & other constraints



Conclusion

"Computing is turning into a utility, and the effects of this transition will ultimately change society as completely as the advent of cheap electricity did."

- The Big Switch by Nicholas Carr

Review Questions ...

Introduction to Cloud Computing -
CET322



Questions?



Thank you!